Association Rule Mining For Retail Transaction Data

1. Introduction

In this report, we explore the application of association rule mining techniques to retail transaction data. Using the Apriori and FP-Growth algorithms, we aim to uncover patterns and associations in customer purchasing behavior.

2. Objectives

- **To identify frequent itemsets in retail transactions.**
- **To generate strong association rules that reveal valuable insights.**
- **To compare the efficiency and effectiveness of Apriori and FP-Growth algorithms.**

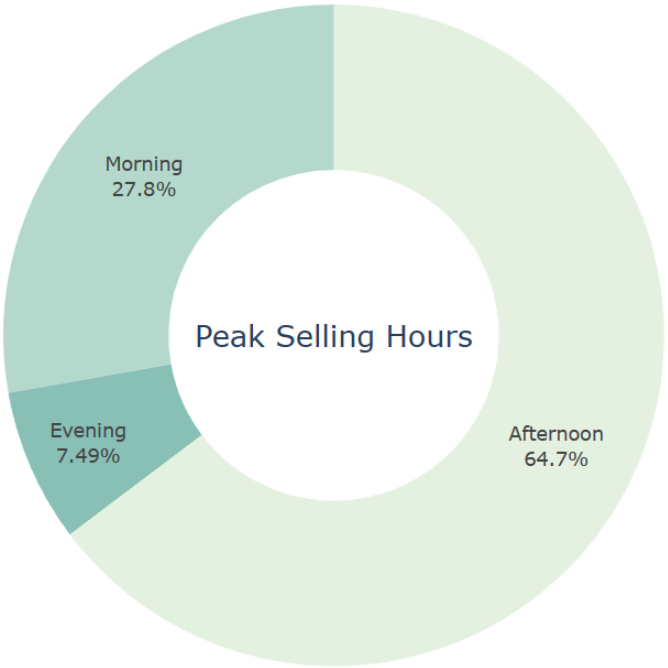
3. Methodology

3.1 The Dataset

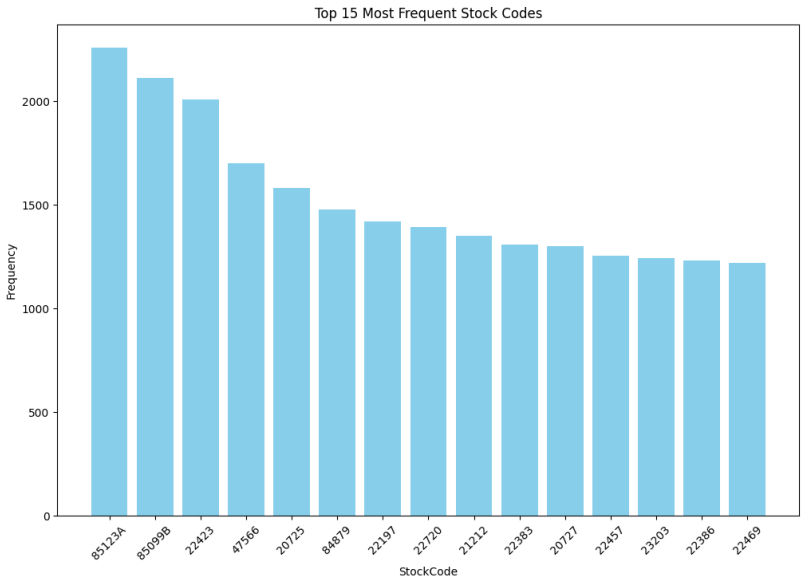
We used the "Online Retail Data Set" available on Kaggle, which contains transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based online retailer.

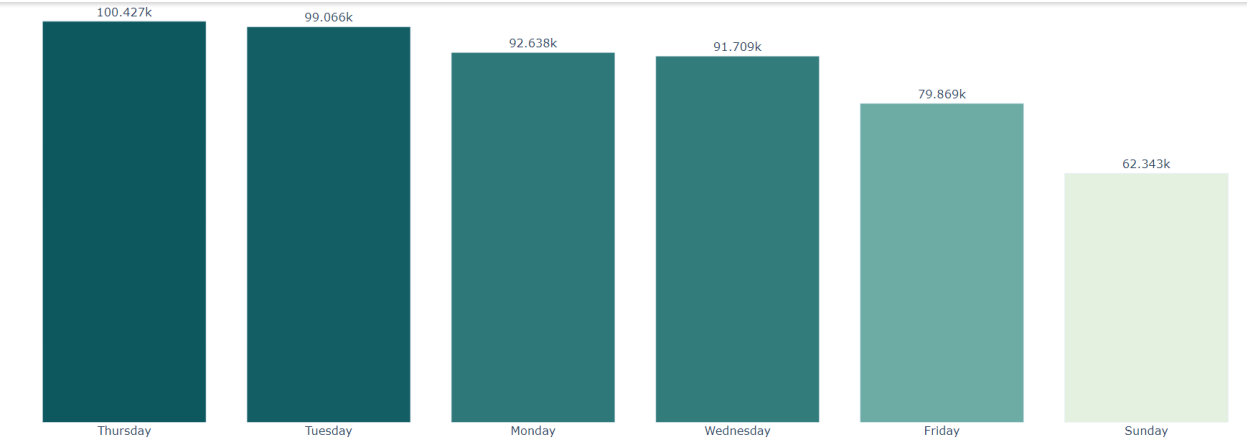
During the EDA phase, several key insights were discovered:

- **Top-Selling Item: The most popular item is the '85123A T-light holder', indicating a high demand for decorative items in the store.**
- **Purchasing Time: Most purchases are made in the afternoon, accounting for 64.7% of total sales. This suggests that customers are more active during this time of the day.**

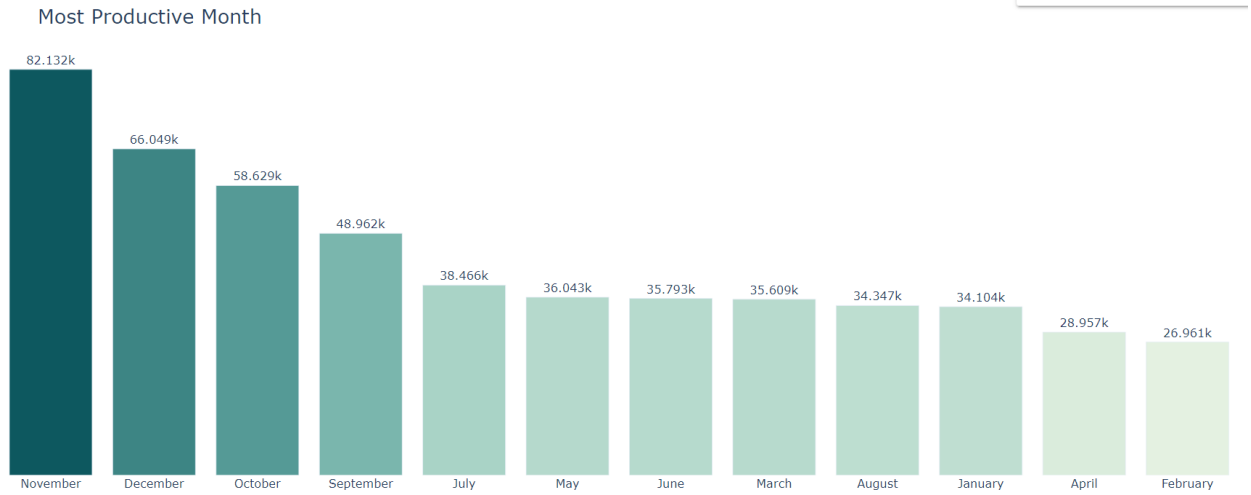


- **Day of the Week: Sales peak on Thursdays, indicating a potential trend where customers prefer shopping towards the end of the workweek.**

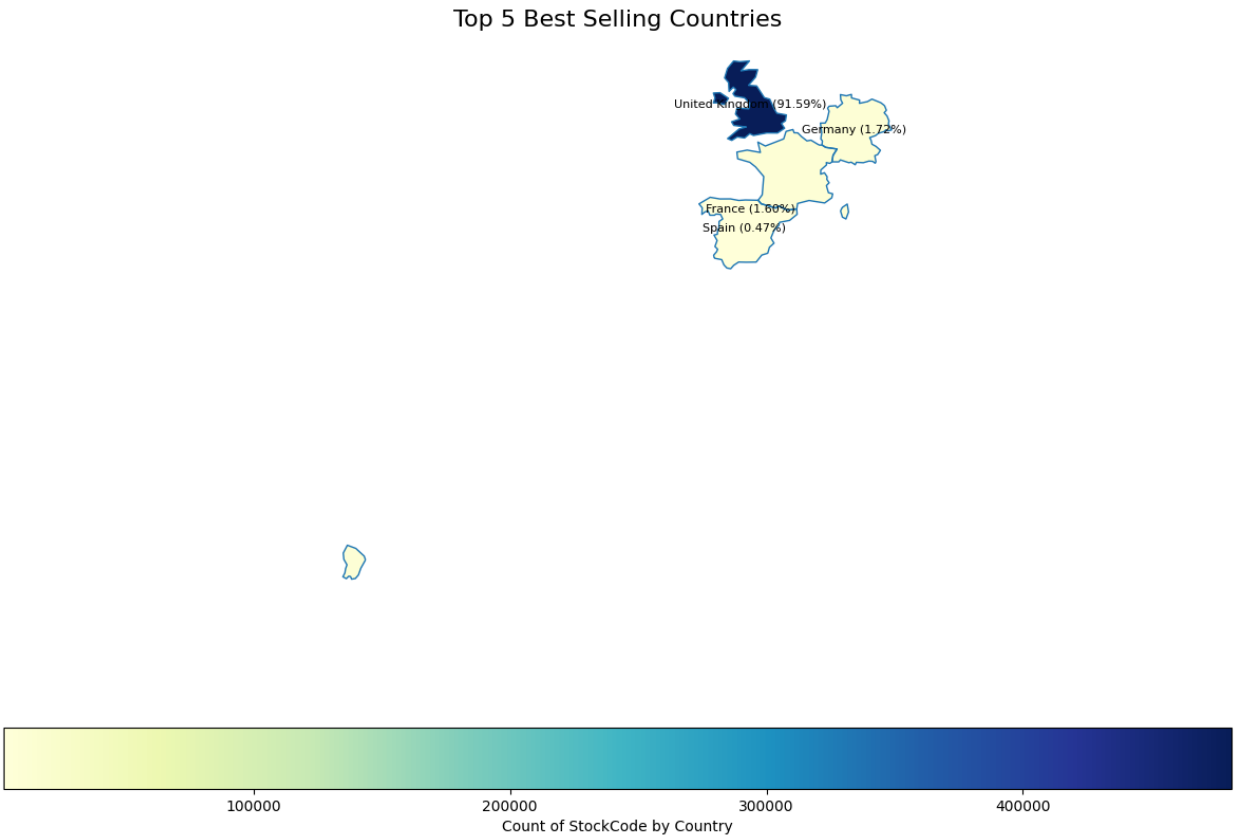




- **Month of the Year: November experiences the highest sales volume, which could be attributed to holiday shopping and seasonal promotions.**



- **Country Analysis: The United Kingdom is the leading country in terms of sales, with 91.59% of the transactions, followed by Germany, France, and EIRE (Ireland). This highlights the store's primary market being the UK, with significant contributions from other European countries.**



3.2 Algorithm 1: Apriori

This method follows a bottom-up approach. We begin with each individual item in the itemset list and then generate candidates through self-joining. The length of the itemsets is extended one item at a time. At each stage, a subset test is conducted, and itemsets containing infrequent subsets are pruned. This process is repeated until no additional frequent itemsets can be derived from the data.

3.3 Algorithm 2: FP-Growth

The FP-tree is a compact representation of the itemset database. This tree structure not only preserves the itemsets in the database but also maintains the relationships between these itemsets. The tree is built by sequentially mapping each itemset to a path in the tree. The underlying principle of this construction is that "items that occur more frequently are more likely to share paths." We then recursively mine the tree to extract frequent patterns. The algorithm, named

pattern growth, generates these frequent patterns by concatenating patterns obtained from conditional FP-trees.

3.4 Measures

1. **Support: Support measures how frequently an itemset appears in the dataset. It is a frequency constraint characterized by downward closure, meaning that all subsets of a frequent set (support > minimum support threshold) are also frequent. This property is used to prune the search space in level-wise algorithms like the APRIORI algorithm, as no superset of an infrequent set can be frequent. However, support has the disadvantage of the rare item problem, where infrequent but potentially valuable items are pruned.**
2. **Confidence: Confidence measures how often a rule has been found to be true. It is not downward closed and was developed alongside support in the support-confidence framework. While support prunes the search space to leave potentially interesting rules, confidence filters rules that exceed a minimum confidence threshold. However, confidence is sensitive to the frequency of the consequent (Y) in the dataset. Higher support for Y will automatically produce higher confidence values, even if there is no actual association between the items.**
3. **Lift: Lift is the ratio of the observed support to that expected if X and Y were independent. It measures the difference between X and Y appearing together in the dataset versus what would be expected if they were independent. In a sales setting, it helps determine how many more units of items X and Y together are sold than expected from independent sales. Using minimum leverage thresholds incorporates an implicit frequency constraint. For example, setting a minimum leverage threshold to 0.01% (equivalent to 10 occurrences in a dataset with 100,000 transactions) first involves finding all itemsets with a minimum support of 0.01%, then filtering these itemsets using the leverage constraint. Like support, leverage can also suffer from the rare item problem.**

4. Results

Comparison of Apriori and FP-Growth Algorithms

In this section, we compare the results of the Apriori and FP-Growth algorithms using different minimum support thresholds. The comparison focuses on the number of frequent itemsets identified, the execution time, and the quality of the association rules generated.

Parameters Used

- **Apriori Algorithm:**
    - **Minimum Support: 0.02**
    - **Minimum Lift: 1.0**
- **FP-Growth Algorithm:**
    - **Minimum Support: 0.01**
    - **Minimum Lift: 1.0**

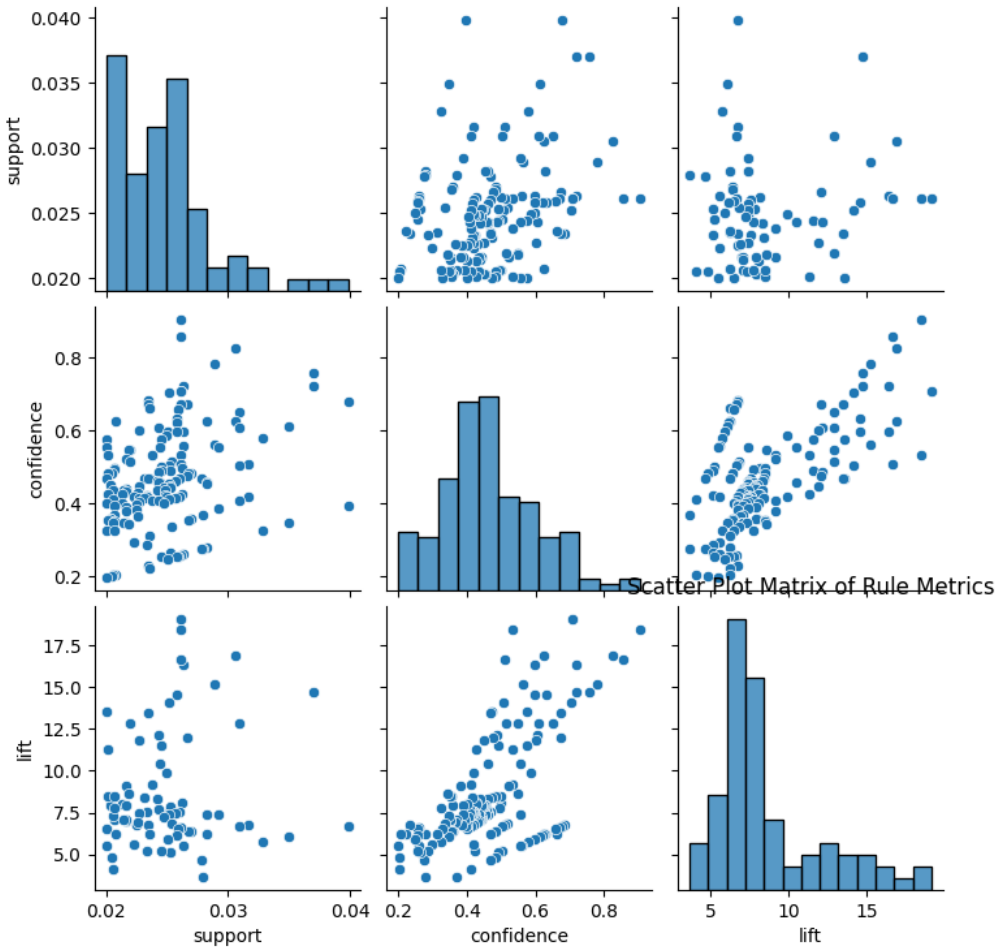
The reason for not using the same minimum support value for both algorithms is due to the Apriori algorithm's inefficiency with large datasets. When using a minimum support of 0.2 for both algorithms, the results were similar. However, by reducing the minimum support to 0.1 for the FP-Growth algorithm, we observed a significant increase in the number of rules and higher confidence levels. Specifically, the FP-Growth algorithm produced over 3000 rules with a highest confidence of 0.95, compared to the Apriori algorithm which generated over 100 rules with a highest confidence of 0.9 using a minimum support of 0.2.

Pairplot Visualization

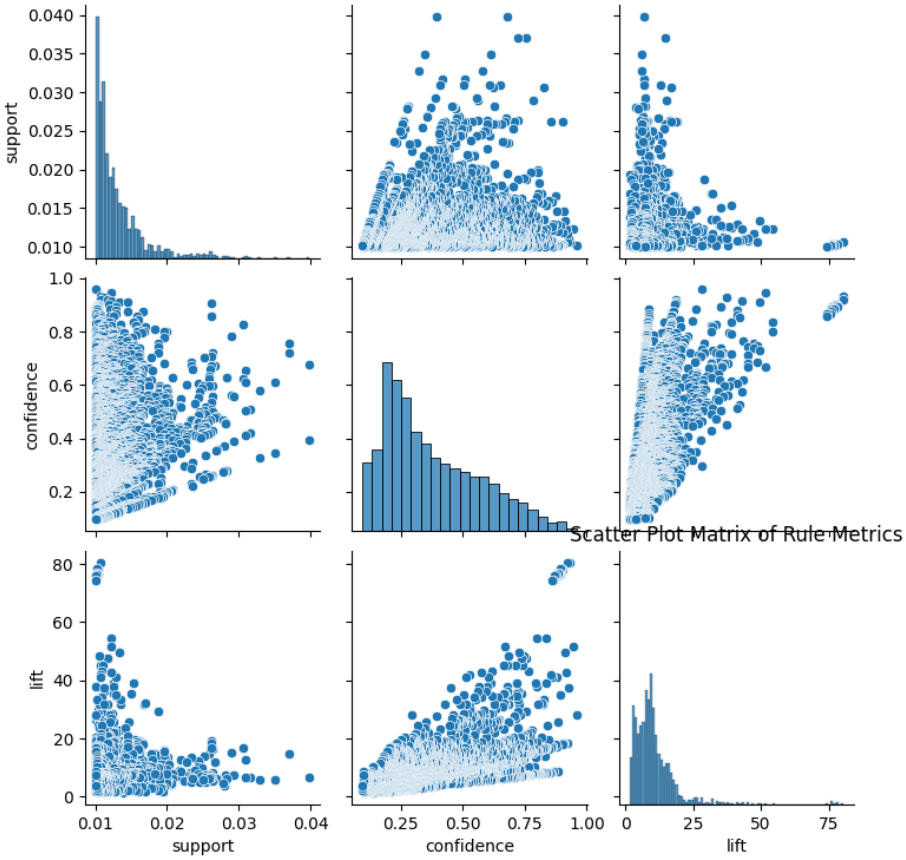
A pairplot was used to visualize the relationships between different measures of the association rules such as support, confidence, and lift. Pairplots are particularly useful for visualizing the distribution of data and detecting potential correlations between variables.

- **The pairplot revealed that higher confidence values are often associated with higher support values, indicating that more frequently occurring itemsets tend to have higher confidence levels.**
- **The distribution of lift values showed that most of the association rules have a lift value greater than 1, indicating positive associations between itemsets.**

Pairplot for Apriori Algorithm:



Pairplot for FP-Growth Algorithm:



Network Graph Visualization

A network graph was used to visualize the association rules as a network of nodes and edges, where nodes represent items and edges represent the association rules

between them. This type of visualization helps in understanding the complex relationships and interactions between different items in the dataset.

- **The network graph highlighted the most significant association rules, with thicker edges representing higher confidence levels.**
- **Central nodes in the graph represent items that are frequently associated with many other items, indicating their importance in the dataset.**

The visualizations provided a deeper insight into the structure and strength of the associations discovered by the algorithms, aiding in the interpretation and practical application of the results.

The network graphs are in the Colab notebook

5. Conclusion

The FP-Growth algorithm proved to be more efficient and effective for large datasets, generating a higher number of rules with better confidence levels compared to the Apriori algorithm. The findings from the association rules provide valuable insights into customer purchasing patterns, which can be leveraged for targeted marketing and inventory management.

6. References

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